

Chapter 6 — EDA workflow and tools

Exploration needs a repeatable sequence and a tool stack

Learning objectives

- Pass

Step 1 — Data understanding

- Review structure with head, dtypes, and missing or duplicate flags
- Know which columns are numeric, categorical, or temporal before summarizing
- Example 6.1 begins here on the e-commerce customer table

Step 2 — Descriptive statistics

- Compute means, medians, modes, variance, and standard deviation for numerics
- `Describe()` in pandas bundles many numeric summaries

Step 3 — Visualization

- Apply histograms, box plots, scatter plots, and bar charts from earlier parts
- Visualization reveals patterns summaries alone miss and supports stakeholder communication

Step 4 — Data quality checks

- Use Section 6.4 as a checklist, repair in Chapter 5, then re-explore
- Iteration is normal: a fixed table may expose new relationship questions

Tools — pandas, matplotlib, seaborn

- Pandas handles tables, `read_csv`, `describe`, `isnull`, `drop_duplicates`
- Matplotlib provides base plotting
- R users mirror many steps with `dplyr` and `ggplot2` mentioned in the chapter

Automating EDA

- Libraries such as pandas-profiling or ydata-profiling generate summary reports quickly
- Automation starts exploration; it does not replace domain questions

Takeaways

- Follow understand, summarize, visualize, quality-check in order
- Pandas plus seaborn cover most chapter workflows
- Automate the first pass, then drill into anomalies manually

Next

- Complete the quiz for this part
- Continue to advanced EDA techniques including correlation depth